Sehyun CHOI

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EDUCATION

The Hong Kong University of Science and Technology (HKUST) — **Hong Kong**
Master of Philosophy (MPhil) in Computer Science, supervised by Prof. Yangqiu Song — 09/2022 - 09/2024
- **CGA**: 4.12/4.30 (Asian Future Leaders Scholarship Awardee)

Bachelor of Engineering (BEng) in Computer Science, Minor in Bioengineering — 09/2017 - 06/2022
- **GGA**: 4.01/4.30 (First Class Honours; Dean's List and Academic Achievement Medal Awardee)
- **Relevant Coursework**: Machine Learning for NLP, Deep Learning in Computer Vision, Big Data Management, Honours Software Engineering, Honours Algorithms, Honours Object Oriented Programming and Data Structures, Linear Algebra

Handong International School — **South Korea**
Advanced Placement (AP) by College Board and ACT — 02/2011 - 02/2016
- **Subject**: Physics I (5/5), Physics II (5/5), Calculus BC (5/5) | ACT (34/36)

PROFESSIONAL & RESEARCH EXPERIENCE

TwelveLabs — **South Korea**
ML Research Scientist — 10/2024 - Present
- Implemented video-native frame selection and patch tokenization technology, effectively reducing the maximum token budget by 50% while increasing the average benchmark score by 5%p.
- Developed an effective research consolidation protocol with Multi-teacher On-Policy Distillation, enabling parallel research work and integration into a single release weight.
- Improved video-LLM training efficiency 2x by improving distributed training communications, video token load balancing, and incorporating & writing cutting-edge GPU kernels.
- Improved Video-LLM on-policy RL efficiency 4x for video inputs with overlapped communication, node-aware rollout routing, and efficient multimodal cache scheduling and communications.

NucleusAI — **U.S., Remote**
Research Intern — 09/2023 - 07/2024
- Implemented from scratch a high-inference-speed language model based on the Retentive Network architecture and pretrained it with scalable training techniques such as zero-redundant data parallelism, pipeline parallelism, and tensor parallelism across multi-GPU and multi-node systems.

Naver Corporation — **South Korea**
Machine Learning Research Intern (Papago) — 12/2021 - 02/2022
- Experimented with Deep Active Learning approaches for querying the most critical data points from unlabeled datasets in Computer Vision and Natural Language Understanding for Machine Translation (MT) tasks.
- Improved fine-tuning efficiency by selecting informative data based on corpus statistics, pretrained MT Transformer output uncertainty, and the model's sentence encoder representations.
- Identified uncertainty, representativeness, and diversity as essential aspects of Active Learning data selection, and devised metrics to quantify them.

KAIST Artificial Intelligence (AI) Lab — **South Korea**
Summer Research Intern — 07/2021 - 08/2021
- Supported a language model detoxification project by researching eXplainable AI methods to detect potentially offensive concepts captured in language models and ablate them.
- Proposed a novel framework for detecting offensive concepts in language models from non-offensive sentences.

Skelter Labs — **South Korea**
Machine Learning Engineer Intern — 12/2019 - 07/2020
- Conducted experiments to improve deep learning-based speech synthesis models, including introducing new network modules, using data augmentation, and fine-tuning on different datasets.
- Used data pipelines, microservices, and cloud services such as Spark, Kubernetes, and GCP to run large-scale training jobs.
- Participated in collaborative software development using Git and code review systems.

PUBLICATIONS

KCTS: Knowledge-Constrained Tree Search Decoding with Token-Level Hallucination Detection
EMNLP 2023 Main Conference | arxiv.org/abs/2310.09044

AbsPyramid: Benchmarking the Abstraction Ability of Language Models with a Unified Entailment Graph
Arxiv Preprint | arxiv.org/abs/2311.09174

CKBP v2: An Expert-Annotated Evaluation Set for Commonsense Knowledge Base Population
Arxiv Preprint | arxiv.org/abs/2304.10392

Benchmarking Commonsense Knowledge Base Population with an Effective Evaluation Dataset
EMNLP 2021 Main Conference | DOI: 10.18653/v1/2021.emnlp-main.705

AWARD & CERTIFICATE

Naver Clova AI Rush 2021 & 2022 — **South Korea**
2nd place in Unknown Document Classification (2022) and Smart Grammar Editor (2021) — 2021, 2022
- Trained Korean pretrained language models with model calibration and representation-distance-based out-of-distribution detection to identify unknown document distributions while maintaining in-domain classification performance.
- Improved the benchmark GLEU score by 33% over the vanilla Transformer baseline for Korean Grammatical Error Correction (GEC) by implementing pseudo-data creation with back-translation, two-stage pretraining, and data mixing during fine-tuning.

SKILL & INTEREST

Language: Korean (Native) | English (Fluent) | Python
Interest: NLP | LLM | Controllable Generation | Knowledge Grounding & Reasoning | AI Safety | XAI